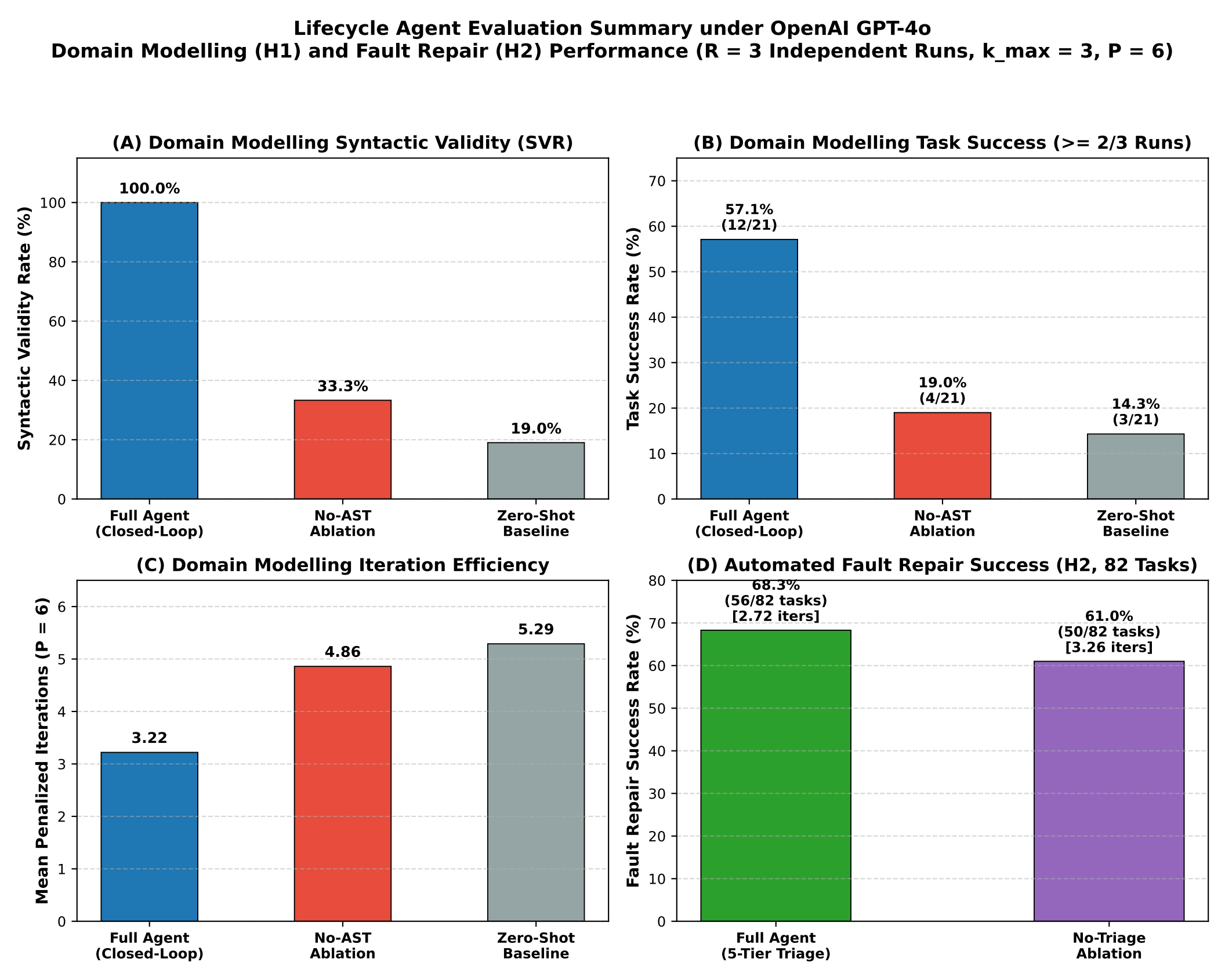
Lifecycle Agent Evaluation Summary under OpenAI GPT-4o
Domain Modelling (H1) and Fault Repair (H2) Performance (R = 3 Independent Runs, k_max = 3, P = 6)
(A) Domain Modelling Syntactic Validity (SVR)
Syntactic Validity Rate (%)
100.0%
33.3%
19.0%
Full Agent (Closed-Loop)
No-AST Ablation
Zero-Shot Baseline
(B) Domain Modelling Task Success (>= 2/3 Runs)
Task Success Rate (%)
57.1% (12/21)
19.0% (4/21)
14.3% (3/21)
Full Agent (Closed-Loop)
No-AST Ablation
Zero-Shot Baseline
(C) Domain Modelling Iteration Efficiency
Mean Penalized Iterations (P = 6)
3.22
4.86
5.29
Full Agent (Closed-Loop)
No-AST Ablation
Zero-Shot Baseline
(D) Automated Fault Repair Success (H2, 82 Tasks)
Fault Repair Success Rate (%)
68.3% (56/82 tasks) [2.72 iters]
61.0% (50/82 tasks) [3.26 iters]
Full Agent (5-Tier Triage)
No-Triage Ablation